

Reproduction of DeReFusion and Diagnostics of Operator Specialization for Financial Time Series

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Abstract—This report consolidates the work completed up to 2026-09-12 on the reproduction of DeReFusion and on a sequence of controlled diagnostics that ask a single question: does the structural state of the input determine which operator is more useful? We first reproduce the core comparison of the original paper — a DLinear base branch, an LSTM–Transformer residual branch and parameter-free additive fusion, all under RevIN — and then stratify forecast errors by a causal relative-volatility measure. The marginal value of the nonlinear branch turns out to be *asset-dependent* rather than universal: significantly positive and seed-robust on the S&P 500, directionally positive but statistically inconclusive for ETHUSD, and significantly *reversed* for BTCUSD. A capacity-sensitivity experiment (hidden width $23 \rightarrow 128$, all else frozen) shows that the earlier “linear dominates everywhere” reading was partly an artefact of a narrow nonlinear bottleneck: at adequate capacity ETHUSD prefers the nonlinear operator in 36/36 observations while BTCUSD never does. A capacity-matched proxy routing experiment (twelve pre-registered structural states, linear versus MLP, identical protocol) finds no robust state-level operator specialization. Finally, an exploratory sweep over ten assets yields interaction effects that span zero (six negative, four positive, five supported) and two leave-one-out-stable associations (realized volatility and $|\text{ACF}_1|$), reported as a *candidate* asset-level structural regularity only. We state explicitly what the evidence excludes, what it supports, and what remains unresolved.

Index Terms—reproduction, time series forecasting, operator specialization, structural states, adaptive fusion, controlled comparison.

I. SCOPE AND RESEARCH QUESTIONS

The work follows an evidence-first protocol: reproduce the reference result, then ask one question at a time with every grouping rule, threshold and operator definition fixed *before* outcomes are inspected. Three questions structure the report: **Q-A** does the published ordering of fusion strategies reproduce on independently fetched data; **Q-B** is the marginal contribution of the nonlinear branch conditioned on the volatility regime of the input window; and **Q-C** does the *structural state* of the input determine which operator is more useful, i.e. is there an experimental basis for routing? All experiments run on CPU under a frozen 70/10/20 chronological split and report paired bootstrap intervals (4000 resamples) with paired win

TABLE I
REPRODUCTION OF THE CORE COMPARISON (GSPC, SEED 2021, NORMALIZED SCALE).

Model	$T=1$	$T=24$	R^2	Params
DeReFusion (additive)	0.01647	0.06230	0.8691	28,436
revin-DLinear (linear)	0.01577	0.07007	0.8528	4,664
gatev1 (volatility)	—	0.06767	0.8578	28,580
gatev2 (learnable)	0.02526	0.07191	0.8489	~28k

rates. No hyper-parameter was tuned after observing test outcomes.

II. REPRODUCTION OF THE REFERENCE COMPARISON

The dataset of the original repository is not redistributable, so ten daily OHLC series (2016–2025) were re-fetched; row counts match the paper (2513 for indices and equities, 2602 for FX, 3653 for BTCUSD, 2975 for ETHUSD). Table I reports the S&P 500 comparison at $T \in \{1, 24\}$ (seed 2021).

At $T = 24$ the published ordering reproduces: additive fusion beats the linear branch by 11.1% in MSE, while both gate variants are worse than parameter-free addition. At $T = 1$ the linear branch is marginally better, consistent with the behaviour of long-horizon linear forecasters. Extending the protocol to two further assets at $T = 24$ gives 0.21797 versus 0.22322 for BTCUSD (+2.4%) and 0.14602 versus 0.14876 for ETHUSD (+1.8%); the additive advantage is ordered $\text{GSPC} \gg \text{BTCUSD} \approx \text{ETHUSD}$. Figure 1 shows the corresponding per-horizon prediction curves and the run dashboard of the full model.

III. VOLATILITY-REGIME STRATIFICATION

Each test window is assigned a regime from *causal* volatility measures computed on the input window only: the standard deviation of log returns and a de-trended version, $RV_t^{rel} = RV_t / \text{median}(RV_{t-120:t-1})$. Table II reports the relative improvement of the nonlinear branch per stratum (relative volatility, the primary measure) and the interaction effect $\Delta_{high50} - \Delta_{low50}$.

Two findings follow. The S&P 500 effect is *seed-robust*: both seeds give significantly negative interactions of the same order, and the calm strata are indistinguishable between branches. BTCUSD reverses the sign significantly,

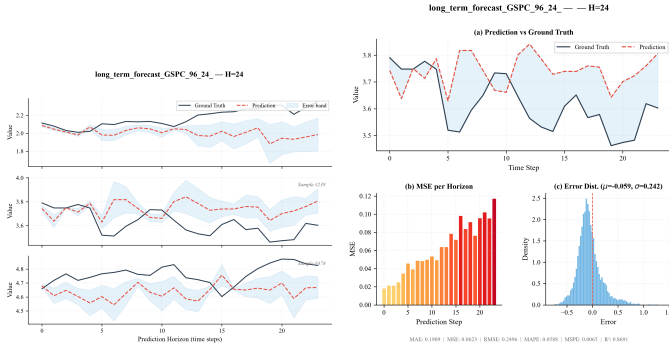


Fig. 1. Reproduction of DeReFusion on GSPC ($T=24$): forecast curves (left) and the per-run diagnostic dashboard (right).

TABLE II

STRATIFIED PERFORMANCE, RELATIVE VOLATILITY ($\Delta = \text{MSE}_{\text{nonlin}} - \text{MSE}_{\text{lin}}$; NEGATIVE FAVOURS THE NONLINEAR BRANCH).
[†]CI EXCLUDES ZERO.

Asset	low 20%	high 50%	high 20%	Interaction
GSPC (2021)	+6.0%	+14.2% [†]	+21.6% [†]	-0.01205 [†]
GSPC (2022)	-2.7%	+23.2% [†]	+29.6% [†]	-0.02248 [†]
ETHUSD	+16.7% [†]	+6.3% [†]	+14.2% [†]	-0.00943
BTCUSD	+3.5%	-1.0%	-3.5%	+0.01502 [†]

so “nonlinear advantage grows with volatility” cannot be stated as a universal criterion. A methodological check explains part of the earlier confusion: raw volatility is strongly confounded with time (correlation of RV with the sample index: +0.44 for GSPC, -0.67 for BTCUSD). The de-trended RV^{rel} removes that trend, and under it the spurious “linear wins in the middle quintile” of GSPC disappears; all reported figures therefore use RV^{rel} .

IV. ADAPTIVE FUSION DOES NOT PAY

The natural use of the regime structure would be a gate $\alpha(X)$ mixing the two branches, and the framework already provides exactly this variant. On the asset where the structure is strongest and seed-robust (GSPC, $T = 24$) the volatility-aware gate obtains MSE 0.06767, i.e. 8.6% worse than parameter-free addition (0.06230) and only 3.4% better than the plain linear branch (0.07007). Because the nonlinear branch is never worse than the linear one in the calm strata, the optimal gate weight is essentially constant at one, leaving no switching headroom. Together with the previously reported negative results for the learnable gate, this motivates stopping the adaptive-fusion line rather than building a router on top of it.

V. CAPACITY SENSITIVITY

The proxy experiment of Section VI compares a linear operator with a capacity-matched MLP (9,240 versus 9,431 parameters, +2.1%). That matching forces a 384-dimensional input through a 23-dimensional hidden layer, a severe bottleneck. We therefore re-ran the structural-state evaluation with the hidden width as the *only* variable

TABLE III

CAPACITY SENSITIVITY. MEAN ΔMSE (NEGATIVE = NONLINEAR BETTER); IN PARENTHESES, THE NUMBER OF THE 36 (STATE, SEED) OBSERVATIONS PREFERRING THE NONLINEAR OPERATOR.

Hidden	Params	GSPC	BTCUSD	ETHUSD
23	9,431	+0.0598 (0/36)	+0.1783 (0/36)	+0.0145 (12/36)
64	26,200	+0.0250 (7/36)	+0.1888 (0/36)	-0.0281 (36/36)
128	52,376	+0.0061 (8/36)	+0.2998 (0/36)	-0.0321 (36/36)

(23 $\rightarrow$ 64 $\rightarrow$ 128; data, features, splits, optimizer, learning rate, batch size, epochs, early stopping, seeds and evaluation protocol unchanged). Table III reports the mean ΔMSE over twelve structural states and three seeds.

Three pre-registered checks agree. (i) *Sign reversals*: 25 state-level reversals across widths, 23 of them ETHUSD (all seeds) and 8 GSPC (all at seed 2023 only, i.e. seed-unstable); BTCUSD has none. (ii) *Spread across states*: the range of ΔMSE across states *shrinks* with capacity for GSPC (0.106 $\rightarrow$ 0.060) and ETHUSD (0.089 $\rightarrow$ 0.040); the BTCUSD increase is driven by a single high-error state. (iii) *Per-seed direction consistency* falls from 31/36 to 29/36 to 28/36. The ETHUSD preference is *asset-wide* once capacity is sufficient — all twelve states reverse, not a subset — so it is asset-level heterogeneity rather than state-level specialization. We therefore record the earlier uniform “linear dominates” statement as partly capacity-driven, and do not treat the capacity sweep as routing evidence.

VI. STRUCTURE-AWARE ROUTING: A MINIMAL CONTROLLED PROXY

A. Design

Twelve structural states were defined *before* any outcome was inspected, using causal window features: autocorrelation at lags 1/5/10, a trend t -statistic, sign persistence, a jump ratio with a fixed threshold of $c = 3\sigma$, the de-trended relative volatility, skewness and kurtosis. Binary states use train-set medians as thresholds; a composite *structure strength* state and a $K=3$ k -means cross-check are also reported. The two operators are a linear map (9,240 parameters) and an MLP with one 23-unit hidden layer (9,431 parameters), trained under an identical protocol (same seed, optimizer, steps, batch, learning rate, early stopping and preprocessing).

B. Result

At matched capacity the linear operator is better in *every* one of the twelve states for GSPC and BTCUSD (GSPC ΔMSE between +0.047 and +0.081, significant in all three seeds; BTCUSD between +0.129 and +0.448). Only ETHUSD shows between-state variation of both signs (structure-weak states favouring the nonlinear operator by up to -0.013, structure-strong states favouring the linear one by +0.033). No state preference survives the capacity sweep. **Main finding: no robust operator**

TABLE IV
ASSET-LEVEL INTERACTION EFFECTS (RELATIVE VOLATILITY). $\dagger$ CI
EXCLUDES ZERO.

Asset	$\Delta_{interaction}$	95% CI	Supported
EURUSD	-0.1371	[-0.164, -0.111]	yes †
GSPC	-0.0173	[-0.027, -0.009]	yes †
USDJPY	-0.0122	[-0.026, +0.002]	no
ETHUSD	-0.0094	[-0.019, +0.000]	no
DJI	-0.0084	[-0.019, +0.003]	no
TM	-0.0081	[-0.028, +0.011]	no
SOX	+0.0044	[-0.008, +0.017]	no
BTCUSD	+0.0150	[+0.006, +0.024]	yes †
BABA	+0.0405	[+0.022, +0.060]	yes †
NVO	+0.0976	[+0.046, +0.149]	yes †

TABLE V
EXPLORATORY SPEARMAN ASSOCIATION OF ASSET STRUCTURE
FEATURES WITH $\Delta_{interaction}$ ($N = 10$) AND LOO STABILITY.

Feature	ρ	p	LOO range	Stable
ACF ₁	+0.733	0.016	[+0.633, +0.817]	yes
Realized volatility	+0.661	0.038	[+0.533, +0.900]	yes
Kurtosis	+0.345	0.328	[+0.100, +0.533]	yes
Trend persistence	+0.042	0.907	[-0.317, +0.283]	no (flips)
Jump ratio	—	—	—	constant

specialization is established at capacity-matched scale; the one cross-asset invariant is that the two operators are indistinguishable in calm regimes.

VII. ASSET-LEVEL HETEROGENEITY ACROSS TEN ASSETS

The frozen protocol ($T = 24$, seed 2021) was then run on seven additional assets to map the *sign* of the interaction effect. GSPC aggregates its two seeds into a single asset, so $N = 10$. Table IV reports the interaction effect and Table V the exploratory association between asset structure features and that effect, including a leave-one-asset-out (LOO) check.

The sign of the effect spans zero: six assets negative, four positive, five statistically supported. Two features are LOO-stable: assets with higher realized volatility and stronger first-order autocorrelation show a *less* favourable (or reversed) nonlinear advantage in turbulent regimes. At $N=10$ both associations reach $p < 0.05$ ($|ACF_1|$ $\rho = +0.733$, $p = 0.016$; realized volatility $\rho = +0.661$, $p = 0.038$), and the inclusion of SOX (interaction +0.0044, statistically inconclusive) leaves the verdict unchanged. **These associations are exploratory and are reported as a candidate regularity only**: with ten assets a single p -value is not decisive, the jump-ratio feature is constant across assets at the median (one exceedance in 95 returns) and therefore carries no discriminative information at the pre-registered threshold, and trend persistence flips sign under LOO and is marked single-asset sensitive.

TABLE VI
CONSOLIDATED ANSWERS TO THE FIVE DIAGNOSTIC QUESTIONS.

Question	Answer
Q1 Robust sample-level routing evidence?	No. Neither the routing proxy nor the capacity sweep produces a stable state-to-operator mapping.
Q2 Genuine asset-level operator heterogeneity?	Yes. At adequate capacity ETHUSD prefers the nonlinear operator in 36/36 observations, BTCUSD prefers linear at every capacity, GSPC is near parity.
Q3 Explainable by the existing structural features?	Only as a candidate: realized volatility and $ ACF_1 $ associate with the interaction sign and survive LOO, but the sign spans zero across ten assets.
Q4 Principal remaining uncertainty?	Asset dependence and operator form — not the structural representation, and not routing, which is now largely excluded.
Q5 May a stronger nonlinear operator be reconsidered?	Only as one candidate, tested against a capacity-matched MLP, and never by extrapolating “ETHUSD prefers nonlinear” to a specific operator family.

Two further observations qualify the table. First, the four assets whose confidence intervals contain zero (USDJPY, ETHUSD, DJI, TM) are all directionally negative and agree in sign with the supported negative cases, so the negative side is internally consistent even where it is not individually significant. Second, the two supported positive cases (NVO, BABA) are precisely the assets with the strongest first-order autocorrelation, which is what the candidate regularity predicts; the association is therefore not driven by the significance pattern alone but by the ordering of the effect sizes across the feature range.

VIII. DIAGNOSIS AND DECISIONS

The decisions recorded here are: stop the adaptive-fusion line, since a volatility gate measures worse than parameter-free addition and the structure it would exploit leaves no switching headroom; do not build a neural router, because the state-to-operator mapping is not established; treat operator choice as an asset-level configuration question, which is what the evidence supports; and keep any structured nonlinear operator conditional on a reproducible context and on a capacity-matched comparison.

IX. LIMITATIONS

Operator fidelity: the proxy operators (a linear map and an MLP over flattened windows) differ from the framework branches, which combine a DLinear decomposition with an LSTM–Transformer residual under RevIN. Proxy results must therefore not be read as “nonlinearity is useless”;

within the framework the residual branch still outperforms the linear base on the S&P 500 at $T = 24$ across both seeds, and the two statements are consistent. *Sample size*: the asset-level association rests on ten assets and one seed per asset (two for GSPC), so association intervals are wide and the analysis is exploratory. *Regime definition*: regimes are volatility quintiles of a causal window measure, and the calm strata are statistically indistinguishable between operators, so “calm” is an acceptance result rather than a separate effect. *Environment*: runs are CPU-only, so wall-clock times are not comparable to published GPU numbers and only relative comparisons are interpreted.

X. CONCLUSION

The reproduction reproduces the published ordering at $T = 24$ and does so robustly across seeds. The diagnostics that follow refine but do not overturn it: the marginal value of the nonlinear branch is asset-dependent rather than regime-universal, the volatility gate does not pay, and matched-capacity comparisons show that the earlier uniform preference for the linear operator was partly a capacity artefact. No robust experimental basis for sample-level routing was found, whereas genuine asset-level operator heterogeneity was. The next question the evidence supports is therefore not how a router should choose, but in which reproducible contexts a stronger nonlinear operator is worth its capacity — to be answered against a capacity-matched baseline. All artifacts reported here (analysis scripts, batch runners, per-run prediction files and stratification outputs) are committed to the public repository, so every number in this report can be regenerated from the committed commands without access to the original authors’ data.

REPRODUCIBILITY NOTES

All reproduction assets live under `reproduction/` in the project repository: analysis scripts under `reproduction/analysis/`, batch runners under `reproduction/batches/`, per-setting results under `reproduction/results/`. Four environment facts cost real time and are recorded for reuse: the data pipeline imports `patool`, `huggingface_hub`, `sktime` and `datasets` unconditionally; `joblib` must be pinned to 1.5.3; CPU runs require `-no_use_gpu`; and killing DataLoader worker processes (`spawn_main`) deadlocks the parent trainer, so a stalled run must be detected by process CPU time rather than by log output.

REFERENCES

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